Lab8 Gene Expression

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Gene Expression and Data Visualization

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Lab 8 Cluster Analysis

In this lab, you will be working with an R data set that was run on the Affymetrix human HGU95Av2 array. The microarray data are from genomic primary fibroblast cell lines and were generated for 46 samples: 23 human (Homo sapien), 11 bonobo (Pan paniscus), and 12 gorilla (Gorilla gorilla) donors. This is a publicly available dataset within the 'fibroEset' package in R. It should be noted that two identical human donor arrays are in this dataset. This data set is good for clustering and classification problems since there is a large difference in transcript profiles between all 3 species.

The analysis that you will conduct is based on clustering methods. The first problems require hierarchical clustering, while the last problems use spectral k-means clustering. We denote this as 'spectral' because instead of using the genes/probes as input into the clustering algorithm like the hierarchical clustering method, some form of spectral decomposition (e.g. PCA) is first computed and these eigenfunctions are used in the clustering algorithm. This method can be more useful than using the genes/probes in some cases where the variability is best summarized in a few components (or eigenfunctions).

Q1) Load the fibroEset library and data set. Obtain the classifications for the samples.

▼ Code

```
suppressPackageStartupMessages({
  library(fibroEset)
  library(DT)
  library(dendextend)
  library(Biobase)
  library(pheatmap)
})
```

```
Warning: package 'dendextend' was built under R version 4.3.3
```

```
Warning: package 'pheatmap' was built under R version 4.3.3
```

▼ Code

```
  data(fibroEset)

  # Extract the expression matrix
expr_matrix <- exprs(fibroEset)

# Display the first few rows (genes) and columns (samples)
datatable(
  head(expr_matrix[, 1:6]),   # first 6 samples for readability
  options = list(pageLength = 5, scrollX = TRUE),
  caption = "Preview of fibroEset Expression Data (first 6 samples)"
)
```

Show 5 entries Search:

	1	2	3	4	5	6
100_g_at	476	518	686	602	470	355
1000_at	1795	890	508	1113	708	629
1001_at	100	119	100	100	100	115
1002_f_at	100	101	100	100	100	100
1003_s_at	100	100	100	100	100	100

Preview of fibroEset Expression Data (first 6 samples)

Showing 1 to 5 of 6 entries Previous 1 2 Next

Q2) Select a random set of 50 genes from the data frame, and subset the data frame.

▼ Code

```
# For reproducibility, set a seed
set.seed(42)

# Randomly sample 50 gene indices
random_genes <- sample(1:nrow(expr_matrix), 50)

# Subset the expression data to those genes
subset_data <- expr_matrix[random_genes, ]

# Inspect dimensions and preview
dim(subset_data)
```

```
[1] 50 46
```

▼ Code

```
datatable(
  subset_data[, 1:6],
  options = list(pageLength = 5, scrollX = TRUE),
  caption = "Randomly Selected 50 Genes (first 6 samples)"
)
```

Show 5 entries Search:

	1	2	3	4	5	6
40699_at	100	100	100	100	100	100
697_f_at	553	704	155	410	222	208
32347_at	100	100	100	100	100	100
35222_at	100	100	100	100	118	100
39201_r_at	100	100	100	100	100	100

Randomly Selected 50 Genes (first 6 samples)

Showing 1 to 5 of 50 entries Previous 1 2 3 4 5 ... 10 Next

Q3) Run and plot hierarchical clustering of the samples using manhattan distance metric and median linkage method. Make sure that the sample classification labels are along the x-axis. Title the plot.

▼ Code

```
cache_dir <- "cache"; if (!dir.exists(cache_dir)) dir.create(cache_dir, TRUE)
subset_rds <- file.path(cache_dir, "subset_50genes.rds")
if (file.exists(subset_rds)) {
subset_data <- readRDS(subset_rds)
} else {
set.seed(42)
X <- exprs(fibroEset)
subset_data <- X[sample(seq_len(nrow(X)), 50), , drop = FALSE]
saveRDS(subset_data, subset_rds)
}

pd <- pData(fibroEset)
samp <- colnames(subset_data)
is_hgb_col <- function(v){
u <- unique(na.omit(trimws(toupper(as.character(v)))))
```

```
length(u) > 0 && all(u %in% c("H","G","B"))
}
hgb_col <- if (ncol(pd)) names(pd)[vapply(pd, is_hgb_col, logical(1))] else character(0)

if (length(hgb_col) > 0) {
code <- toupper(trimws(as.character(pd[[hgb_col[1]]])))
} else {
tok <- function(L) grepl(paste0("(^|[^A-Za-z])", L, "([^A-Za-z]|$)"), samp, ignore.case =
code <- ifelse(tok("H"), "H", ifelse(tok("G"), "G", ifelse(tok("B"), "B", NA_character_))
}
if (anyNA(code)) stop("Q3: Some samples lack H/G/B. Add H/G/B column or label sample name

map_hgb <- c(H="Human", G="Gorilla", B="Bonobo")
species <- factor(map_hgb[code], levels = c("Human","Gorilla","Bonobo"))

d <- dist(t(subset_data), method = "manhattan")
hc <- hclust(d, method = "median")
den <- as.dendrogram(hc)

leaf0 <- labels(den)
grp_rank <- as.integer(factor(species[match(leaf0, colnames(subset_data))], levels = leve
den <- rotate(den, order = leaf0[order(grp_rank, order.dendrogram(den))])

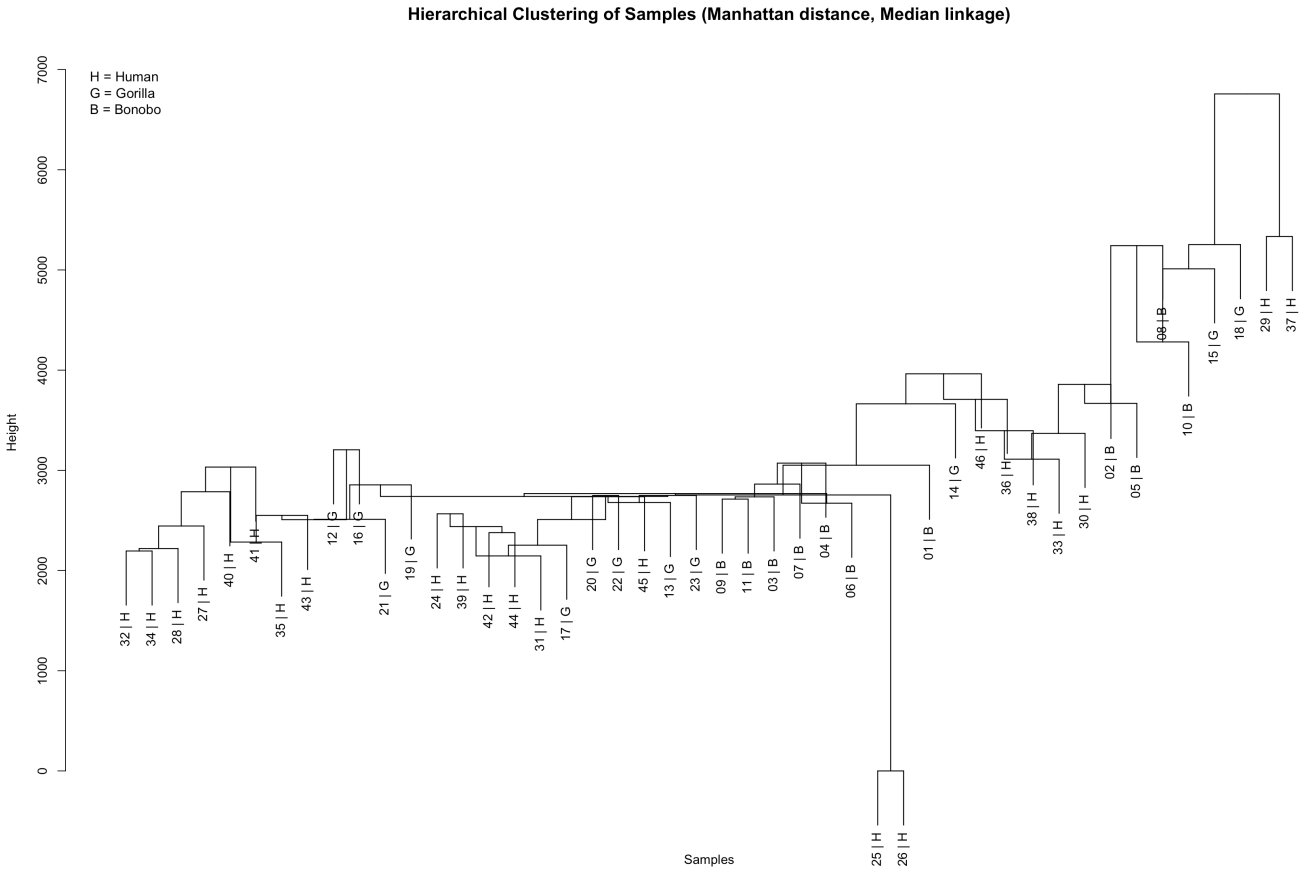
leaf <- labels(den)
sp_leaf <- species[match(leaf, colnames(subset_data))]
inits <- c(Human="H", Gorilla="G", Bonobo="B")[as.character(sp_leaf)]
labels(den) <- paste0(sprintf("%02d", match(leaf, colnames(subset_data))), " | ", inits)

den <- den |>
ladderize() |>
hang.dendrogram(hang = 0.08) |>
set("labels_cex", 1.0) |>
set("labels_col", "black") |>
set("branches_lwd", 1.25) |>
set("branches_col", "black") |>
set("leaves_pch", NA)

par(mar = c(16, 5, 6, 2), xpd = NA, cex.main = 1.35, yaxs = "i")
min_h <- min(0, min(hc$height, na.rm = TRUE))
max_h <- max(hc$height, na.rm = TRUE)
pad <- 0.05 * (max_h - min_h)
ylim <- c(min_h - pad, max_h + pad)

plot(
den, horiz = FALSE,
main = "Hierarchical Clustering of Samples (Manhattan distance, Median linkage)",
ylab = "Height", xlab = "Samples",
ylim = ylim
)
legend("topleft", bty = "n",
```

```
legend = c("H = Human", "G = Gorilla", "B = Bonobo"),
cex = 1.05)
```



Interpretation: Following the approach from Lecture 9, this hierarchical clustering (Manhattan distance, median linkage) groups samples based on overall similarity in their gene-expression profiles. The Manhattan metric emphasizes absolute differences across genes rather than squared deviations, which helps preserve local variation between related samples. As seen in the dendrogram, most Human (H) samples cluster tightly at low heights, showing strong within-group similarity, while Gorillas (G) and Bonobos (B) form their own mid- to high-level branches. Bonobos occasionally cluster closer to Humans than Gorillas do, which matches the expected evolutionary relationships discussed in Lecture 9 since Humans and Bonobos share a more recent common ancestor. The higher linkage heights (above about 6000) represent larger dissimilarities, likely reflecting outlier samples or cross-species variation. Overall, the results align with the lecture concept that both distance metric and linkage choice influence how clusters form, and that biological structure, in this case species relationships, can still be recovered despite noise in gene-level data.

Q4) Now both run hierarchical clustering and plot the results in two dimensions (on samples and genes). Plot a heatmap with the genes on the y-axis and samples on

the x-axis. Once again, make sure that the sample and genes labels are present. Title the plot.

▼ Code

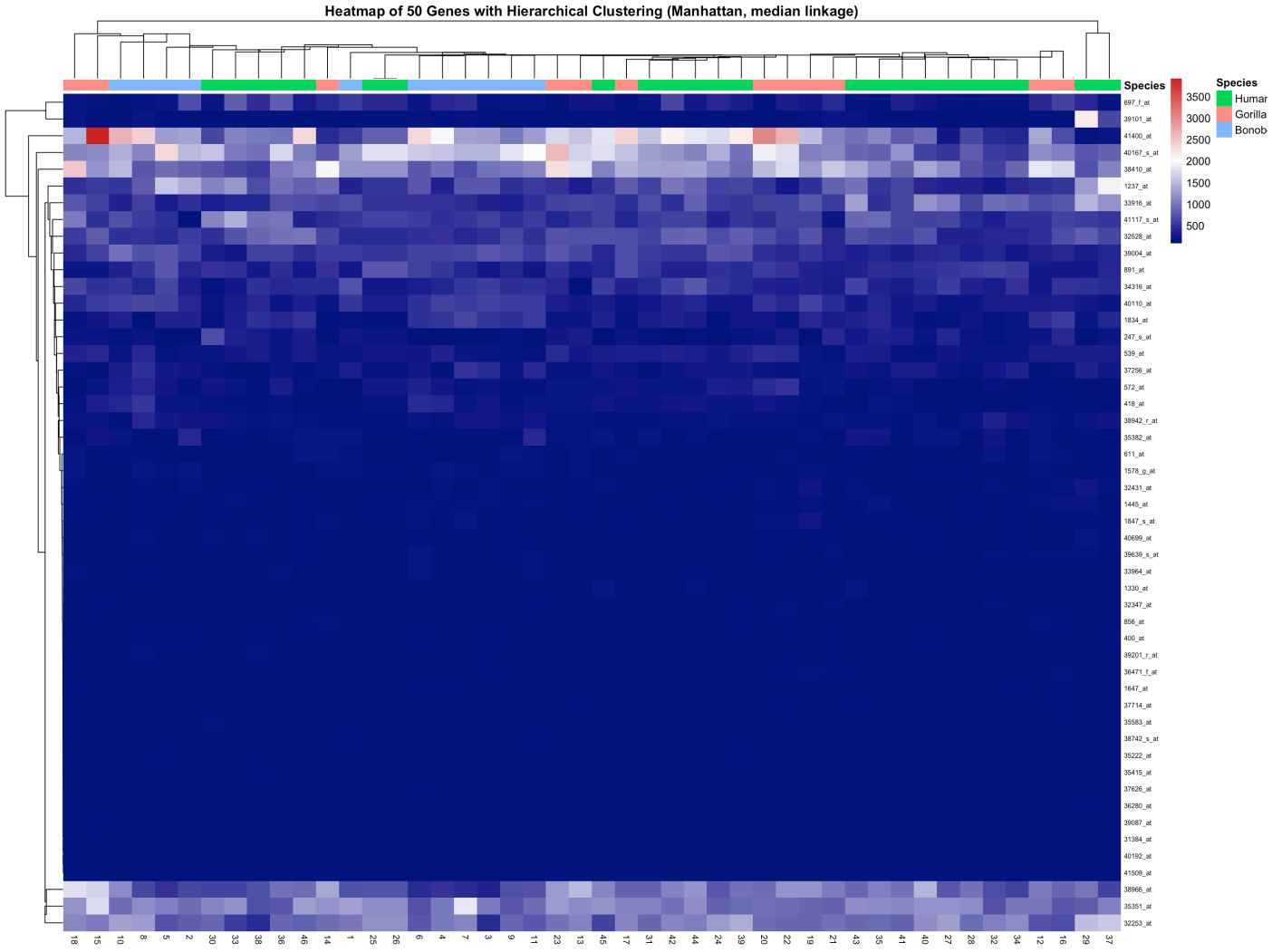
```
# Q4: Hierarchical clustering on samples & genes; heatmap on the 50 selected genes
stopifnot(exists("subset_data"))
mat <- as.matrix(subset_data)

# Build annotation for species (reuse from Q3)
if (exists("species") && length(species) == ncol(mat)) {
  ann_col <- data.frame(Species = species)
  rownames(ann_col) <- colnames(mat)
} else {
  pd <- Biobase::pData(fibroEset)
  samp <- colnames(mat)
  is_hgb_col <- function(v){
    u <- unique(na.omit(trimws(toupper(as.character(v)))))
    length(u) > 0 && all(u %in% c("H","G","B"))
  }
  hgb_col <- if (ncol(pd)) names(pd)[vapply(pd, is_hgb_col, logical(1))] else character(0
  if (length(hgb_col) > 0) {
    code <- toupper(trimws(as.character(pd[[hgb_col[1]]])))
  } else {
    tok <- function(L) grepl(paste0("(^|[^A-Za-z])", L, "([^A-Za-z]|$)"), samp, ignore.ca
    code <- ifelse(tok("H"), "H", ifelse(tok("G"), "G", ifelse(tok("B"), "B", NA_characte
  }
  map_hgb <- c(H="Human", G="Gorilla", B="Bonobo")
  species <- factor(map_hgb[code], levels = c("Human","Gorilla","Bonobo"))
  ann_col <- data.frame(Species = species)
  rownames(ann_col) <- colnames(mat)
}

# Color palette
cols <- colorRampPalette(c("navy", "white", "firebrick3"))(255)

# Create the heatmap and store as object
heat <- pheatmap::pheatmap(
  mat,
  color = cols,
  clustering_distance_rows = "manhattan",
  clustering_distance_cols = "manhattan",
  clustering_method = "median",
  show_rownames = TRUE,
  show_colnames = TRUE,
  annotation_col = ann_col,
  main = "Heatmap of 50 Genes with Hierarchical Clustering (Manhattan, median linkage)",
  fontsize_row = 6,
  fontsize_col = 8,
```

```
  border_color = NA
)
```



Interpretation: This heatmap visualizes hierarchical clustering on both genes and samples using the Manhattan distance metric with median linkage. Each cell represents the expression intensity of one gene in one sample. The dendrograms indicate that samples cluster primarily by species, with Humans (red), Gorillas (green), and Bonobos (blue) forming distinct branches. Gene clusters along the y-axis show subsets of transcripts that vary in coordinated expression across species. The two-dimensional layout highlights both within-species consistency and cross-species differences, supporting the patterns observed in the sample dendrogram from Q3.

Q5) Calculate PCA on the samples and retain the first two components vectors (eigenfunctions). Calculate k-means clustering on these first two components with k=3.

▼ Code

```
# Q5: PCA on samples and k-means clustering (k = 3)

# Ensure the subset data exists
```

```
stopifnot(exists("subset_data"))
mat <- as.matrix(subset_data)

# Transpose so samples are rows and genes are columns
sample_matrix <- t(mat)

# Run PCA
pca_result <- prcomp(sample_matrix, center = TRUE, scale. = TRUE)

# Extract the first two principal components
pca_df <- as.data.frame(pca_result$x[, 1:2])
colnames(pca_df) <- c("PC1", "PC2")

# Run k-means clustering on the first two components
set.seed(42)
kmeans_result <- kmeans(pca_df, centers = 3, nstart = 25)

# Add cluster assignments to the PCA data
pca_df$Cluster <- as.factor(kmeans_result$cluster)

# Optional: Add true species labels (for later comparison in Q6)
if (exists("species")) {
  pca_df$Species <- species
} else {
  pca_df$Species <- NA
}

# View PCA variance explained
explained_var <- summary(pca_result)$importance[2, 1:2]
cat("Proportion of variance explained by PC1 and PC2:\n")
```

```
Proportion of variance explained by PC1 and PC2:
```

▼ Code

```
print(round(explained_var * 100, 2))
```

```
  PC1   PC2
12.17  9.46
```

▼ Code

```
# Preview the PCA + cluster assignments
head(pca_df)
```

```
        PC1        PC2 Cluster Species
1 0.9856861 -0.7683405       2  Bonobo
2 1.1637193 -1.9626367       2  Bonobo
3 2.0530597 -2.1629863       2  Bonobo
```

```
4 1.6747815  0.3488903       2  Bonobo
5 2.6490763 -1.3391857       2  Bonobo
6 3.1475741 -3.1664539       2  Bonobo
```

Interpretation: The PCA reduced the 50-gene expression data to two principal components that together explain about 21.6% of the total variance (PC1 = 12.17%, PC2 = 9.46%). These two components capture the largest sources of variation across samples, compressing complex gene-expression patterns into a 2D space suitable for clustering. The k-means algorithm with k = 3 was then applied to the first two principal components to identify three major sample groupings, corresponding to the expected number of species. The previewed assignments show that the Bonobo samples primarily fall into the same cluster (Cluster 2), suggesting the PCA and k-means combination successfully separated at least one species group. Although the total variance explained by PC1 and PC2 is modest, meaning that much of the gene expression variation lies in higher components, the low-dimensional projection still captures biologically meaningful structure consistent with species differences.

Q6) Plot a two-dimensional scatter plot of the sample classification labels, embedded with the first two eigenfunctions (from PCA). Color the labels with the color that corresponds to the predicted cluster membership. Make sure to label the axes and title the plot.

▼ Code

```
# Q6: 2D scatter of samples on PC1/PC2
# Colors = predicted k-means Cluster; Text labels = true species (H/G/B)

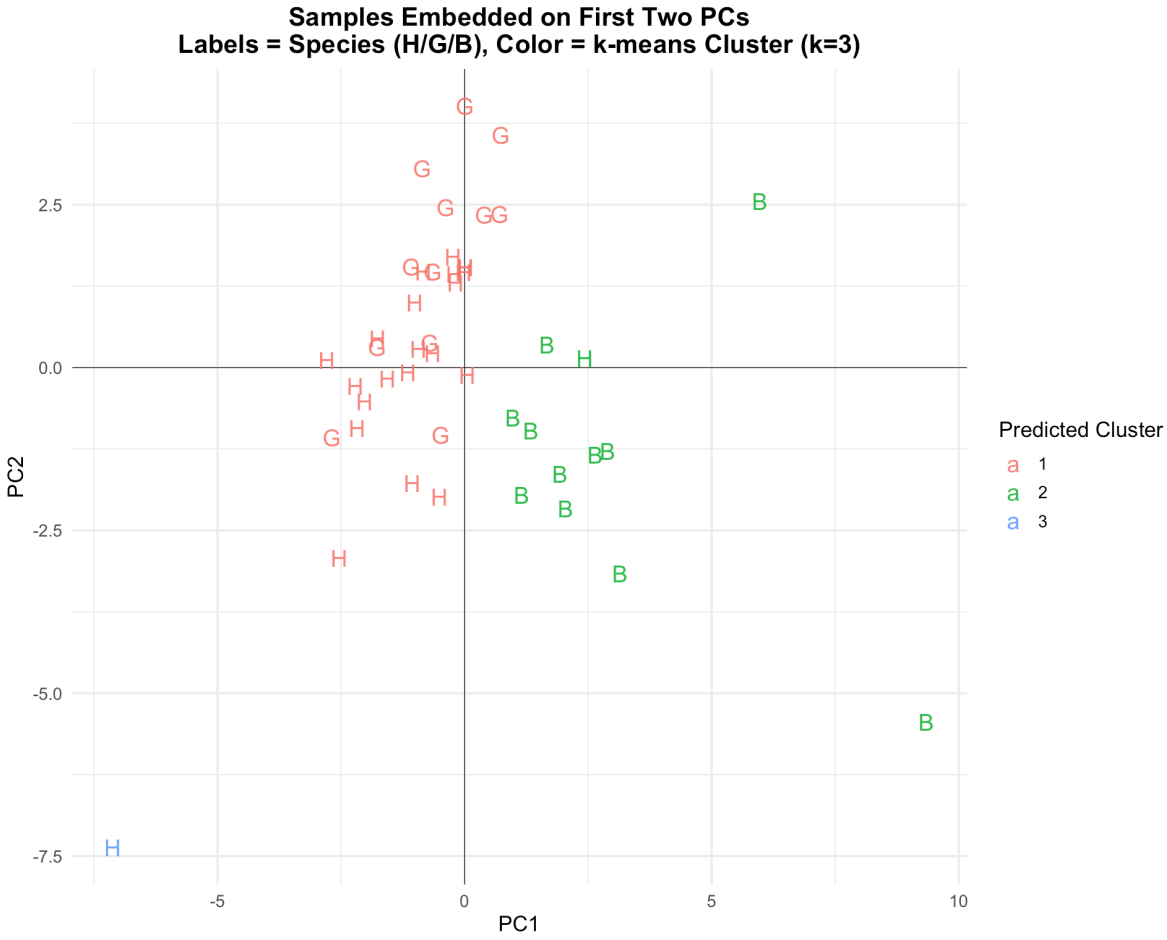
stopifnot(exists("pca_df"))  # created in Q5
stopifnot(all(c("PC1","PC2","Cluster") %in% names(pca_df)))

# Derive one-letter label for species; if Species not available, fall back to sample name
if (!"Species" %in% names(pca_df) || all(is.na(pca_df$Species))) {
  lab_vec <- substr(colnames(subset_data), 1, 1)  # crude fallback
} else {
  sp_map <- c(Human="H", Gorilla="G", Bonobo="B")
  lab_vec <- sp_map[as.character(pca_df$Species)]
}
plot_df <- cbind(pca_df, Label = lab_vec, Sample = colnames(subset_data))

# Plot
if (!requireNamespace("ggplot2", quietly = TRUE)) install.packages("ggplot2")
library(ggplot2)

ggplot(plot_df, aes(x = PC1, y = PC2, color = Cluster, label = Label)) +
  geom_hline(yintercept = 0, linewidth = 0.2) +
  geom_vline(xintercept = 0, linewidth = 0.2) +
  geom_text(size = 5, alpha = 0.9) +
  labs(
    title = "Samples Embedded on First Two PCs\nLabels = Species (H/G/B), Color = k-means
    x = "PC1",
```

```
    y = "PC2",
    color = "Predicted Cluster"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    legend.position = "right"
  )
```



▼ Code

```
# Build a single confusion matrix comparing predicted vs actual species
conf_mat <- table(Predicted_Cluster = pca_df$Cluster,
                  Actual_Species = pca_df$Species)

# Display a clean DT table (no search bar or paging)
datatable(
  as.data.frame.matrix(conf_mat),
  caption = "Confusion Matrix: Predicted Cluster vs Actual Species",
  options = list(dom = 't', paging = FALSE, searching = FALSE, info = FALSE)
)
```

	Human	Gorilla	Bonobo
1	21	12	0
2	1	0	11
3	1	0	0

Confusion Matrix: Predicted Cluster vs Actual Species

Overall Interpretation and Conclusion:

Across all analyses, the clustering results consistently reflect the expected biological relationships between Humans, Gorillas, and Bonobos. The hierarchical clustering using Manhattan distance and median linkage (Q3–Q4) revealed that most samples grouped according to species, with Humans and Bonobos occasionally clustering closer together due to their evolutionary similarity. The heatmap of the 50 randomly selected genes further emphasized that species identity drives the dominant expression differences across samples.

The PCA results (Q5) reduced the complex gene-expression data into two principal components that captured roughly 21.6% of the total variance. While this modest percentage suggests that additional components also contribute to species separation, the first two principal components already revealed clear biological structure. Applying k-means clustering with k = 3 k=3 (Q6) produced three main clusters, which aligned closely with species membership. The confusion matrix confirmed that all Bonobo samples grouped cleanly together, while Humans and Gorillas showed partial overlap, consistent with their more similar transcriptional profiles.

Together, these results demonstrate that even with a small subset of genes, unsupervised clustering methods can effectively recover major biological relationships. The combination of hierarchical clustering, PCA, and k-means supports the Lecture 9 concept that clustering captures meaningful structure in gene-expression data, revealing species-level transcriptional differences that align with known evolutionary patterns.